Answer Key

Quiz 4

Quiz Code: **047530**

Your Performance

15 / 27

56% - Submitted on March 19, 2026 09:25

Quiz Information

Total Questions:	**27**
Time Limit:	**12 minutes**
Quiz Period:	**Mar 19, 2026 09:15 - Mar 19, 2026 11:30**
Answer Key Released:	**March 19, 2026 23:59**

All Questions with Correct Answers & Your Selections

Question 1 ✓ You got this right!

1. What fundamentally distinguishes unsupervised learning from supervised learning?

A It uses labeled target variables

B It requires human intervention

C **It learns from data without predefined labels**
★ Your Answer ✓ Correct Answer

D It only predicts categorical outcomes

Question 2 ✓ You got this right!

2. Which of the following is a primary task of unsupervised learning?

A Predicting stock prices

B **Finding hidden patterns and structures in data**
★ Your Answer ✓ Correct Answer

C Classifying spam emails

D Translating languages

Question 3

3. In unsupervised learning, the algorithm learns exclusively from:

A Target variables

B **Input data features** Correct Answer

C Reward signals

D Pre-classified categories

Question 4

4. What is the main objective of clustering in machine learning?

A To predict a continuous value

B **To group similar data points together** Correct Answer

C To reduce the number of features

D To maximize the number of outliers

Question 5

5. Which of the following is a common real-world application of clustering?

A **Customer segmentation** Correct Answer

B Predicting housing prices

C Detecting objects in an image

D Translating text to speech

Question 6 You got this right!

6. In a highly effective clustering model, intra-cluster distance should be ___ and inter-cluster distance should be ___.

A Maximized; minimized

B Minimized; minimized

C Maximized; maximized

D **Minimized; maximized** Your Answer Correct Answer

Question 7

7. What does the 'K' in the K-means algorithm represent?

A The number of features in the dataset

B **The number of clusters to form** Correct Answer

C The number of iterations allowed

D The learning rate

Question 8

8. Which of the following steps is NOT part of the standard K-means algorithm?

A Initializing cluster centroids

B Assigning points to the nearest centroid

C Updating centroid positions

D **Building a hierarchical tree of data points** Correct Answer

Question 9 You got this right!

9. How does K-means assign a data point to a specific cluster?

A By calculating the correlation coefficient

B **By finding the nearest cluster centroid** Correct Answer Your Answer

C By selecting a cluster at random

D By analyzing the point's density

Question 10

10. What happens to the cluster centroids at each update step of the K-means algorithm?

A They are removed if they are empty

B They move to the origin (0,0)

C **They are updated to the mean of all points assigned to that cluster** Correct Answer

D They are randomly reassigned to new points

Question 11

11. What mathematical objective does the K-means algorithm aim to minimize?

A Cross-entropy loss

B Within-cluster sum of squared errors (Inertia)

Correct Answer

C Mean absolute error

D The distance between different centroids

Question 12 ✓ You got this right!

12. How is the new centroid computationally calculated during the update step?

A By finding the median of the cluster points

B By finding the mode of the cluster points

C **By taking the arithmetic mean of the coordinates of all points in the cluster** ★ Your Answer ✓ Correct Answer

D By finding the point with the maximum distance from the old centroid

Question 13 ✓ You got this right!

13. Which distance metric is most commonly used in standard K-means clustering?

A Manhattan distance

B Cosine similarity

C **Euclidean distance** ★ Your Answer ✓ Correct Answer

D Hamming distance

Question 14

14. If the assignments of data points to clusters stop changing, the K-means algorithm has:

A Diverged

B Overfitted

C Failed

D **Converged** ✓ Correct Answer

Question 15

15. When using standard Euclidean distance, what shape do K-means clusters naturally tend to form?

A **Spherical / Circular** Correct Answer

B Crescent-shaped

C Elongated rectangles

D Complex irregular polygons

Question 16 **You got this right!**

16. If you use Manhattan distance instead of Euclidean distance in K-means, how might the cluster boundaries change?

A They become perfect circles

B **They become more axis-aligned or diamond-shaped** Your Answer Correct Answer

C They form long, curved lines

D They become completely random

Question 17 ✔ You got this right!

17. Changing the distance metric in a clustering algorithm directly alters:

A The number of data points

B **The geometric shape of the resulting clusters**

★ Your Answer ✔ Correct Answer

C The value of K

D The number of features used

Question 18

18. What is a significant drawback of standard K-means clustering?

A It only works on categorical data

B **You must specify the number of clusters (K) beforehand**

✔ Correct Answer

C It is too slow for small datasets

D It requires a massive amount of memory

Question 19 ✓ You got this right!

19. How does the standard K-means algorithm handle outliers?

A It automatically removes them

B It assigns them to a special "outlier" cluster

C **It is highly sensitive to them, and they can severely skew centroid positions** ★ Your Answer ✓ Correct Answer

D It is completely immune to their effects

Question 20 ✓ You got this right!

20. Which statement about K-means convergence is true?

A It is guaranteed to find the global optimum

B It never converges on complex datasets

C **It often converges to a local optimum depending on initialization** ★ Your Answer ✓ Correct Answer

D It requires an infinite number of iterations to converge

Question 21 ✔ You got this right!

21. What primary limitation of standard K-means does the X-means algorithm address?

A Sensitivity to outliers

B **The need to manually specify the optimal number of clusters (K)**

★ Your Answer ✔ Correct Answer

C Slow processing speed

D The inability to process categorical data

Question 22

22. How does X-means mathematically decide whether to split a parent cluster into two child clusters?

A **By using a statistical criterion like the Bayesian Information Criterion (BIC)** Correct Answer

B By checking if the cluster has more than 100 points

C By asking the user for input

D By calculating the silhouette score only

Question 23 You got this right!

23. In the X-means algorithm, what is the typical initial starting point?

A A very high value for K (e.g., K=100)

B **A lower bound for K (e.g., K=1 or K=2)** Correct Answer Your Answer

C Randomly assigning every point as a centroid

D Running K-means++ first

Question 24 You got this right!

24. What is a key computational benefit of X-means over manually running K-means multiple times?

A It uses less memory than a single run of K-means

B **It searches for the best K locally, making it faster than exhaustively evaluating every possible K** Correct Answer

Your Answer

C It eliminates the need for calculating distances entirely

D It guarantees a global optimum without any iterations

Question 25

25. What specific issue in standard K-means does the K-means++ algorithm aim to solve?

A **Poor initial centroid selection leading to bad local optima**

Correct Answer

B The inability to handle non-numerical data

C The requirement to set K beforehand

D Slow convergence during the update step

Question 26 **You got this right!**

26. How does K-means++ select its very first centroid?

A It calculates the mean of the entire dataset

B It selects the data point closest to the origin

C **Uniformly at random from the existing data points**

Your Answer Correct Answer

D It chooses the data point with the highest feature values

Question 27 You got this right!

27. In K-means++, how are subsequent centroids (after the first) chosen?

A Randomly from the remaining points

B **With a probability proportional to their squared distance from the nearest existing centroid** Your Answer Correct Answer

C By picking the points closest to the first centroid

D By selecting points at fixed intervals in the dataset